Sarsengaliyev Damir

• Portfolio • sarsengalievdamir3@gmail.com • linkedin • +77475442806 • LeetCode • GitHub

Education

Nazarbayev University Astana, Kazakhstan
Bachelor of Science, Computer Science. GPA 3.61 / 4, ranked 9th of 255 (top 3.5%) Graduated June 2025
Dean's List: Spring 2023, Fall 2024, Spring 2025
Relevant Coursework: Machine Learning: Theory and Practice, Deep Learning, Generative AI, Database Systems, Probability, Artificial Intelligence, Algorithms

Research Experience

Nazarbayev University Astana, Kazakhstan
Research Assistant, part-time (with Prof. Andrey Tkachenko) June 2024 – August 2026

- Built the audio processing pipeline for a study of 7,324 telephone interviews from five waves of Russian public opinion polls, testing whether voice and response behavior reveal concealed political attitudes.
- Performed automatic transcription and timestamped segmentation with Whisper, and separated interviewer and respondent speech with speaker diarization (Pyannote, WhisperX).
- Integrated a Qwen-2.5-3B LLM to clean and normalize noisy speech transcripts for downstream analysis.
- Co-authored a working paper prepared for the 2026 APSA Annual Meeting (see Working Papers).

Nazarbayev University Astana, Kazakhstan
Research Assistant (with Prof. Adnan Yazici) September 2024 – May 2025

- Developed a multimodal emotion recognition model integrating video, audio, and text modalities using deep neural architectures (ConvNeXt for vision, RoBERTa for text, VGG16 for audio features).
- Achieved over 90% accuracy on the IEMOCAP benchmark.
- Deployed the trained model as a real-time video-based emotion classification system with a web interface.

Working Papers

Tarasenko, G., Tkachenko, A., & Sarsengaliyev, D. *"Measuring Sensitivity Bias with Audio Data: Evidence from Wartime Russia."* Draft prepared for the APSA Annual Meeting, 2026.

Professional Experience

Freedom Holding Corp. Almaty, Kazakhstan
Data Scientist August 2026 – Present

- Built product matching, normalization, and clustering pipelines for categories, names, and attributes across millions of products; deployed to the production admin backend service.
- Developed CatBoost models for service-adoption and churn prediction across the company's ecosystem services (ticketing, e-grocery, travel) on a base of millions of users.
- Built a survival model (XGBoost AFT) to predict next-purchase timing, calibrated using users' service-click behavior.
- Developing a customer-insight layer that feeds model predictions into a RAG-backed LLM with a user-sequence encoder to generate natural-language customer profiles.

Smart Grades LLP Astana, Kazakhstan
Software Engineer (Full Stack), part-time December 2024 – June 2026

- Designed and maintained Java Spring Boot microservices and REST APIs with PostgreSQL for a quasi-governmental platform, including services for downstream analytics and model integration.
- Modeled and automated CRM business process workflows in BPMN using Camunda.
- Built Angular web interfaces and improved reliability and data consistency across the distributed system.

Financial Monitoring Agency of the Republic of Kazakhstan Astana
Intern (Software Engineer) February 2024 – May 2024

- Built Python pipelines to collect and structure open-source (OSINT) crypto data for anti-money laundering (AML) investigations.
- Cleaned and structured large-scale web data into datasets for ML model training and investigative analysis.

Digital Economy Development Center LLP / Kazakhtelecom JSC Astana
Intern (Data Engineer) September 2023 – October 2023

- Built data processing pipelines for millions of national product records (MongoDB) and automated computer-vision validation of product labeling.

Skills

Programming Languages: Python, SQL, Java, C++
Statistics & Data Analysis: Hypothesis Testing, Time-Series Analysis (Granger Causality), Experimental Design, Large-Scale Data Processing (DuckDB)
Machine Learning: Supervised & Unsupervised Learning, Gradient Boosting (XGBoost, LightGBM, CatBoost), Clustering & Dimensionality Reduction (K-Means, PCA, UMAP, t-SNE), Model Evaluation & Tuning
NLP, Speech & LLMs: Hugging Face Transformers, RAG, FAISS, Neo4j, Whisper/WhisperX, Pyannote
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, SciPy
Engineering & Tools: Java Spring Boot, Camunda/BPMN, Angular, PostgreSQL, REST APIs, Git/GitHub
Languages: English (IELTS 7.0), Russian (Fluent), Kazakh (Fluent)

Technical Projects

Agentic Legal GraphRAG Pipeline June 2026

- Built an agentic GraphRAG system over 1,000+ legal documents, a ReAct-style tool-calling agent (Groq function calling) that autonomously selects tools across hybrid retrieval, Neo4j graph traversal, and topic lookup to answer multi-hop legal questions (e.g. tracing amendment/repeal chains across laws).
- Hybrid retrieval: multilingual dense embeddings + BM25 sparse retrieval fused via Reciprocal Rank Fusion, with mode-aware cross-encoder reranking across two language indexes.

Prediction Market Calibration Analysis April 2026

- Analyzed 679,643 Kalshi contracts and 72M trades to study how public attention affects prediction market calibration error using DuckDB for large-scale processing.
- Applied K-Means clustering with PCA, UMAP, and t-SNE to identify five distinct market taxonomies; validated via Kruskal-Wallis and Dunn post-hoc tests.
- Conducted Granger causality analysis between Google Trends scores and miscalibration; built FAISS similarity search index over 679K contracts for calibration prediction.

Household Income and Tenure Prediction (AHS 2023) April 2026

- Processed U.S. Census Bureau AHS 2023 dataset (48,527 units, 1,000+ variables) with structured leakage removal, missing value handling, and encoding across mixed feature types.
- Trained and compared five models (Linear/Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost); XGBoost achieved R^2=0.334 for income regression and AUC=0.974 for tenure classification.
- Conducted ablation study quantifying marginal value of person demographics; applied RandomizedSearchCV tuning reducing Random Forest overfitting gap from 0.61 to 0.08.

Extracurricular Activities

Member, German Club (Nazarbayev University) 2022 – 2024

- Attended weekly German language sessions and helped organize cultural events.

Volunteer, 9th Annual NU Student Awards Ceremony 2022

References available upon request.